Compulsory Assignment 2

Group 97 - Predictive Analytics (KAN-CDSCO1005U)

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Part 1

1.1 Time Series Plot, ACF, and Decomposition

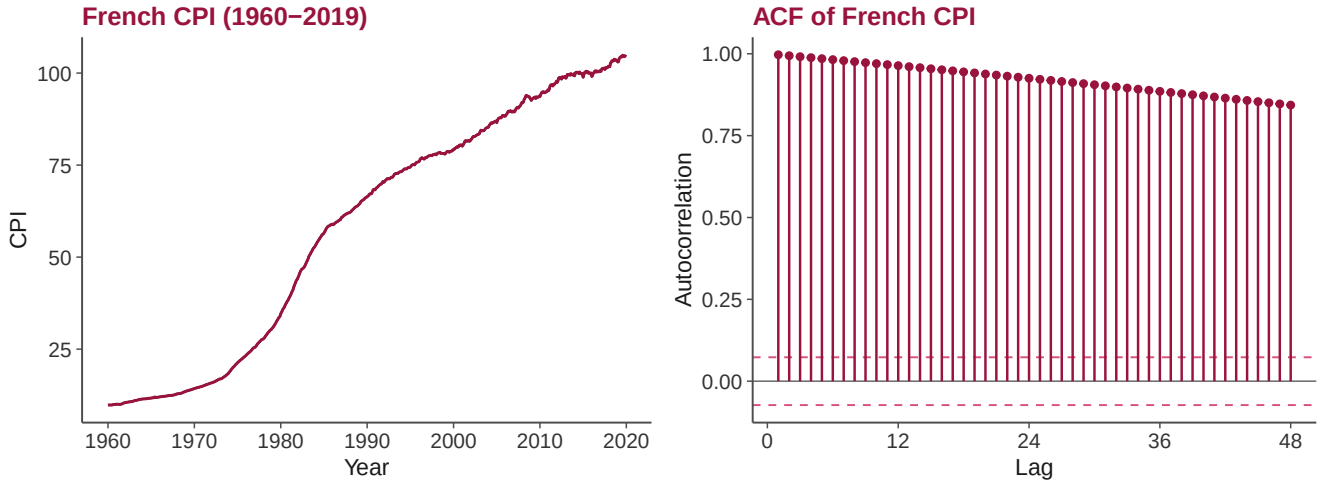

Figure 1: Left: French CPI from January 1960 to December 2019. Right: sample ACF up to 48 lags.

The CPI series displays a strong upward trend with no mean reversion. Growth is steeper in the 1970s–80s (oil crises) and flattens from the mid-1980s onward. The variance increases with the level, suggesting a variance-stabilising transformation is needed. The ACF decays extremely slowly, confirming the series is non-stationary and must be differenced before ARIMA modelling. We apply a classical multiplicative decomposition because the seasonal amplitude grows with the level of the series. This method provides a straightforward separation of trend, seasonal, and irregular components as a first exploratory step.

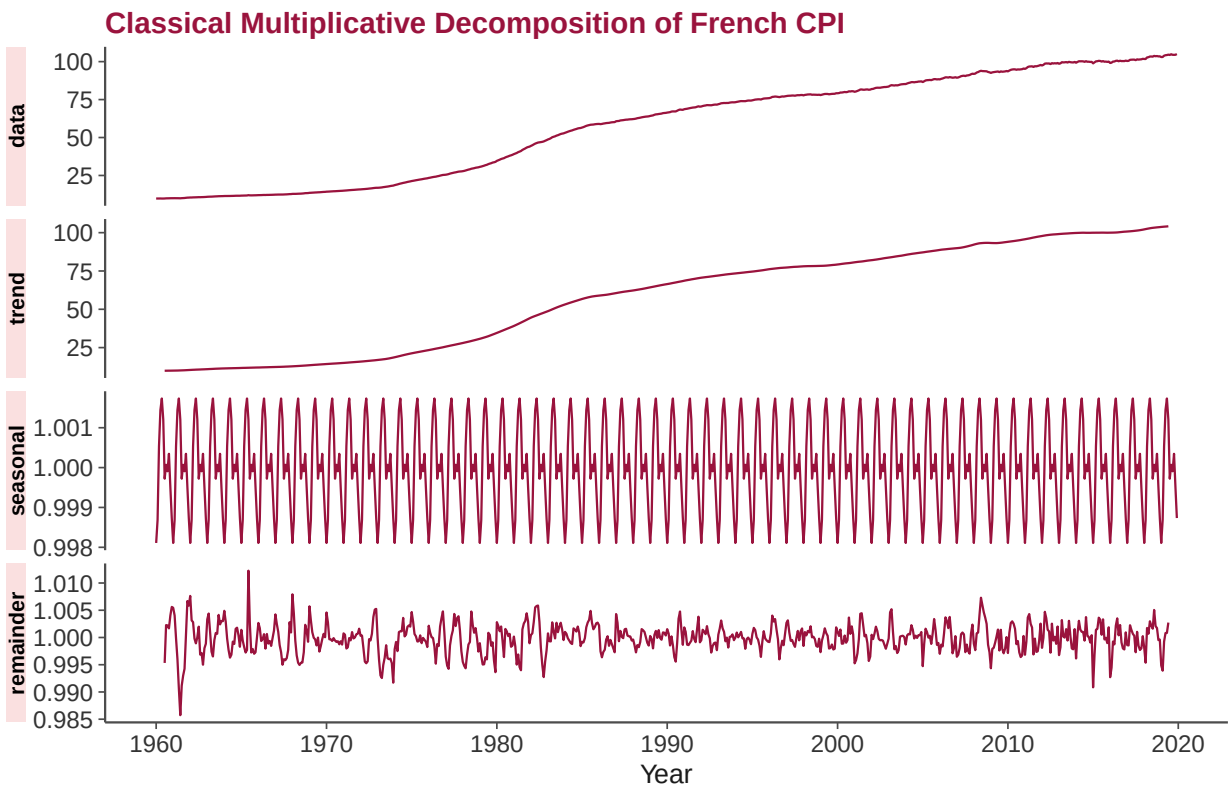

Figure 2: Multiplicative decomposition into data, trend, seasonal, and remainder.

The decomposition reveals a smooth upward trend, a regular annual seasonal pattern, and a remainder

that is largest during the 1970s inflation. The method assumes a constant seasonal pattern and uses symmetric moving averages (losing the first/last six months), which are its main limitations over a 60-year span with structural change.

1.2 Transformation

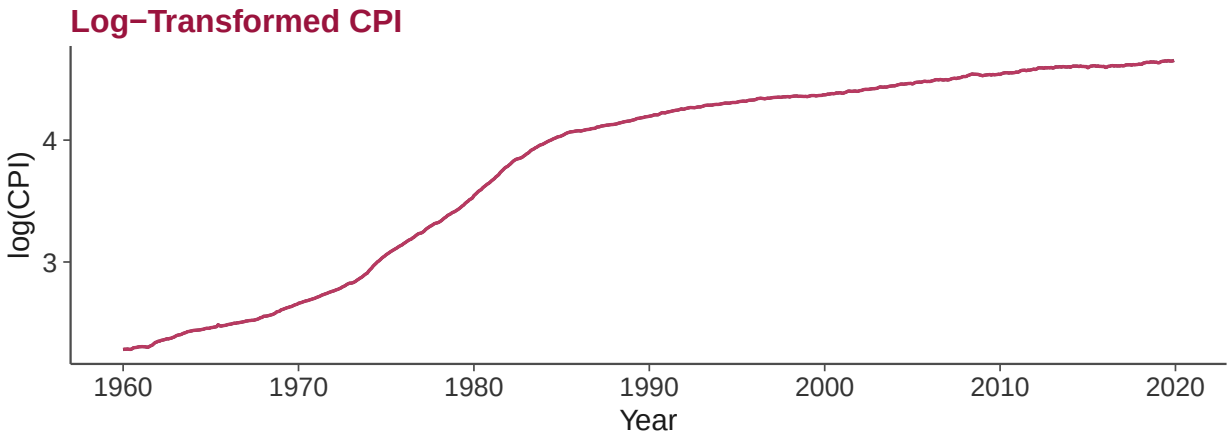

Figure 3: Log-transformed French CPI series.

The Box-Cox method yields $\hat{\lambda} = 1.32$, suggesting a power transformation close to the identity ($\lambda = 1$). However, the raw series exhibits clear variance that grows with the level, and the log transformation produces stable variance with approximately constant seasonal amplitude, making an additive decomposition appropriate. The log-transformed series is used for the remainder of the assignment because first differences of log(CPI) approximate percentage changes, aiding economic interpretation.

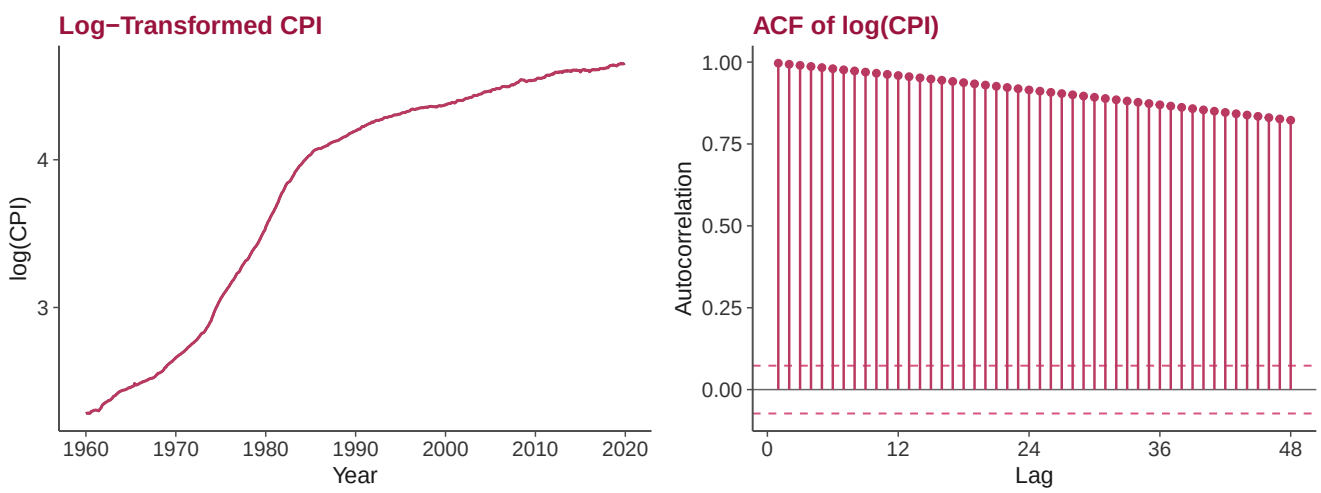

Figure 3b: Left: log-transformed CPI. Right: sample ACF of log(CPI).

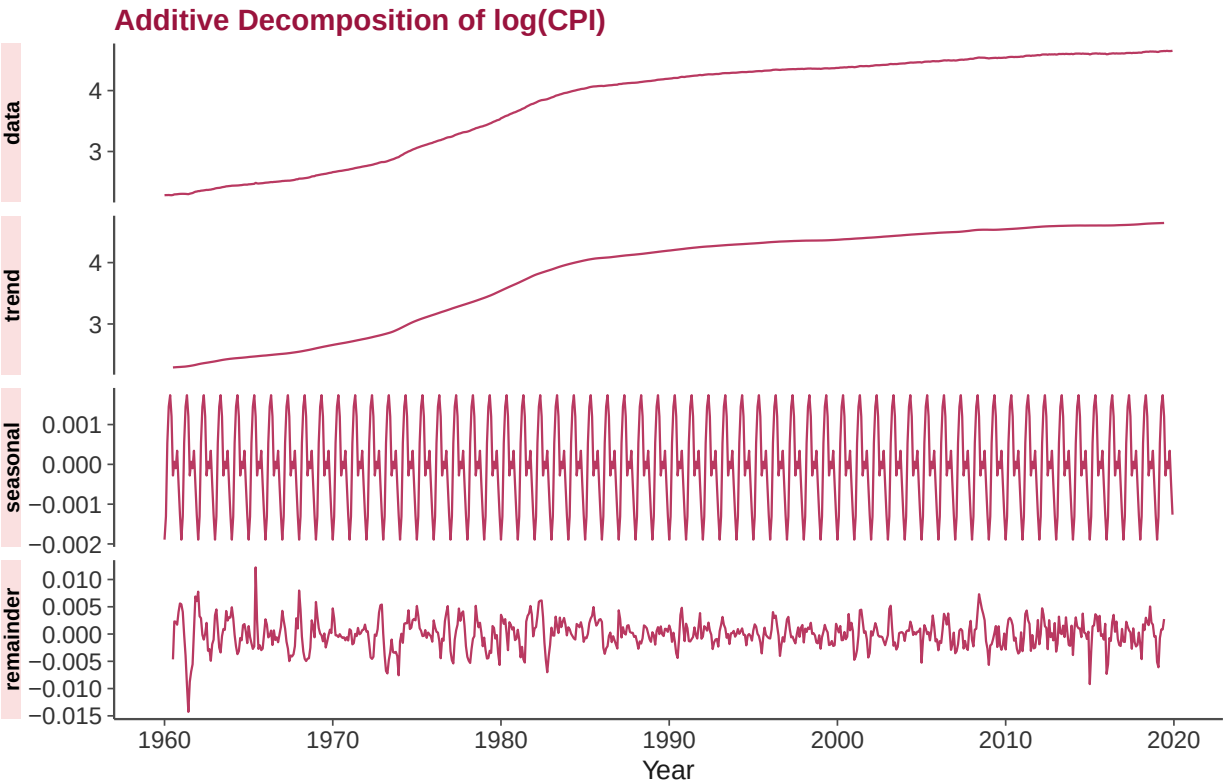

Figure 3c: Additive decomposition of log(CPI) into trend, seasonal, and remainder.

The ACF of log(CPI) still decays very slowly, confirming non-stationarity persists after the transformation. The key difference is in the decomposition: the additive model is now appropriate because the log transform has stabilised the seasonal amplitude. The seasonal component is constant over time, unlike the growing pattern seen in Figure 2. The remainder is also more homogeneous, with reduced magnitude during the 1970s inflation period.

1.3 Unit Root Tests (KPSS and ADF)

The log-transformed CPI has a clear deterministic trend, so we use KPSS with `type = "Trend"` and ADF with `k = 12` (monthly lag structure) to test for stationarity.

Table 1: Unit root tests on log(CPI).

Test	Test Statistic	p-value
KPSS (Trend)	2.4629	0.01
ADF	-1.8044	0.661

KPSS rejects trend-stationarity ($p < 0.01$) and ADF fails to reject the unit root null ($p = 0.661$). Both tests consistently indicate log(CPI) contains a unit root.

1.4 Making the Series Stationary and ARIMA Identification

We apply first-order regular differencing ($d = 1$) and first-order seasonal differencing ($D = 1$, lag 12) to remove the stochastic trend and seasonality.

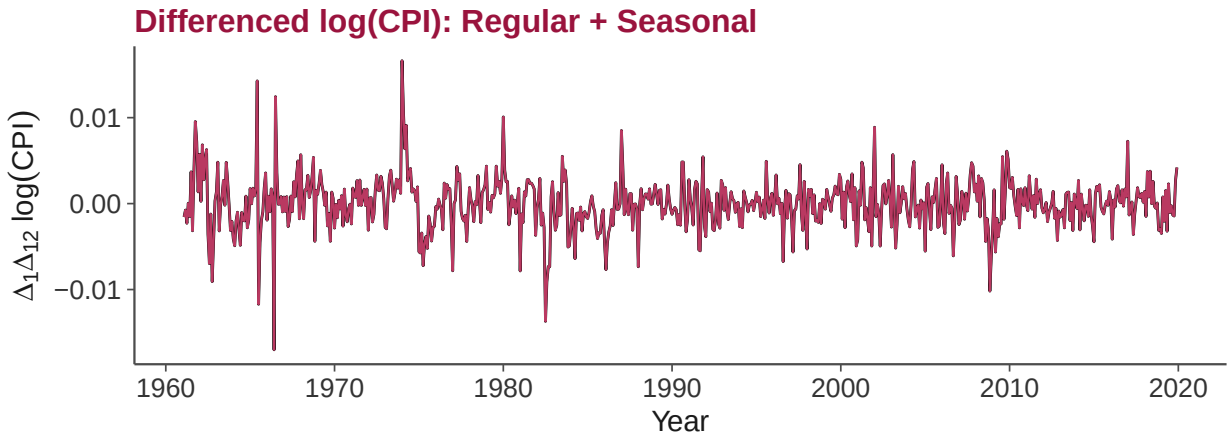

Figure 4: First regular and first seasonal difference of log(CPI).

The KPSS test on the differenced series gives a statistic of 0.1268 ($p = 0.1$), so level stationarity is not rejected.

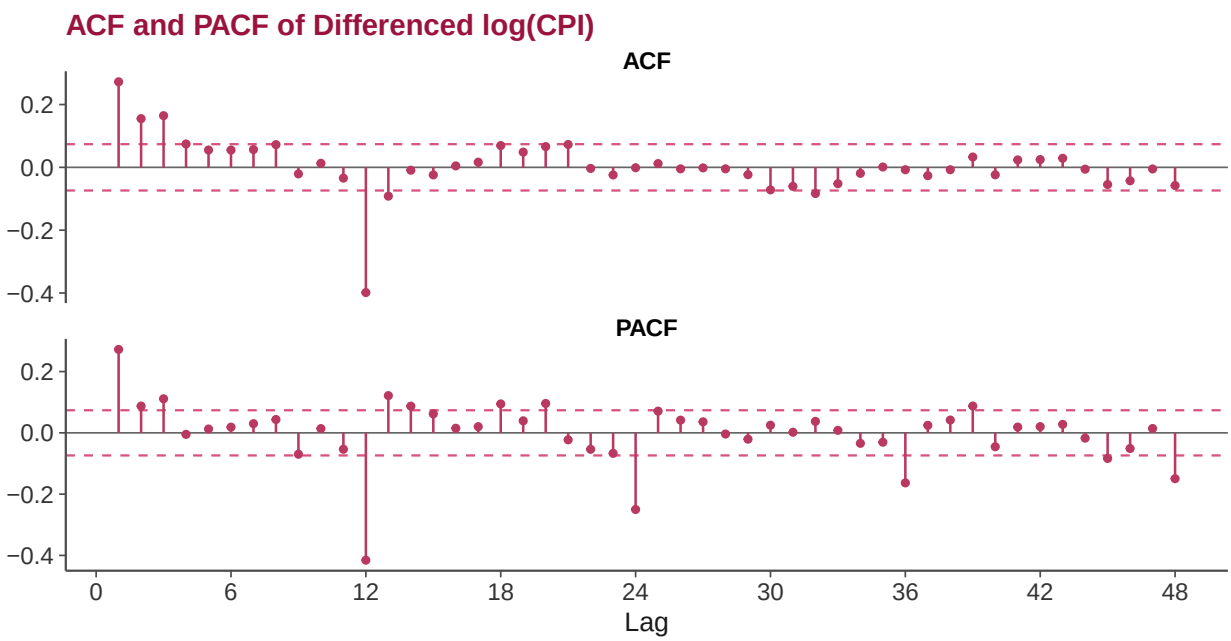

Figure 5: ACF (top) and PACF (bottom) of the doubly differenced log(CPI).

The ACF shows a significant negative spike at lag 1 that cuts off, suggesting a non-seasonal MA(1) component. At seasonal lag 12, a significant negative spike appears, suggesting a seasonal MA(1) component. The PACF tails off at both regular and seasonal lags, consistent with this MA interpretation. Following the identification rules (ACF cuts off at lag q for MA processes, PACF tails off), the proposed model is the "airline model": ARIMA$(0,1,1)(0,1,1)_{12}$, written as $(1-B)(1-B^{12})\log Y_t = (1+\theta_1 B)(1+\Theta_1 B^{12})\varepsilon_t$.

Part 2

2.1 Train/Test Split and Auto ARIMA

The last 24 months (Jan 2018–Dec 2019) are reserved as the test set (24 obs), leaving 696 observations for training.

Using `auto.arima` with exhaustive AIC search, the selected model is ARIMA(1,2,2)(2,0,0)[12]. This differs from the manual specification, suggesting that additional AR and seasonal AR terms improve

in-sample AIC.

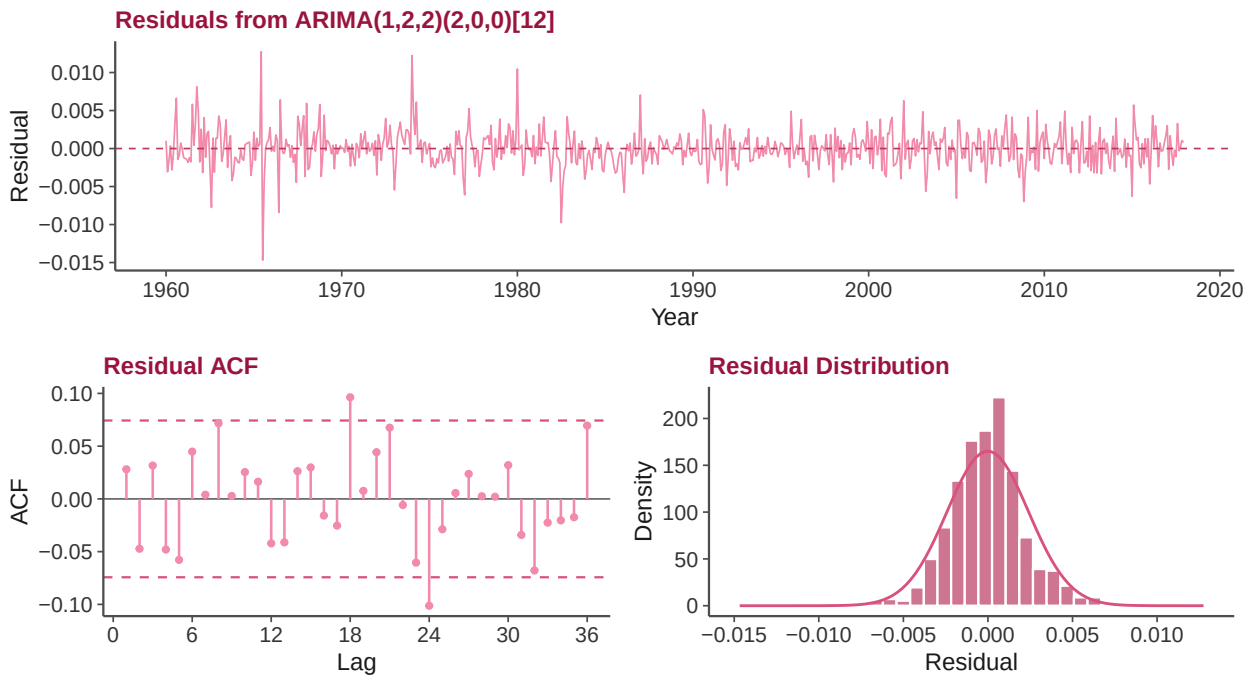

Figure 6: Residual diagnostics. Top: time series. Bottom-left: ACF. Bottom-right: histogram with normal overlay.

The Ljung-Box test at 24 lags gives $Q^* = 38.18$ ($p = 0.00563$), indicating some residual autocorrelation at the 5% level. The largest deviations concentrate in the high-inflation 1960s–70s. The histogram is approximately normal but slightly heavy-tailed. The model could be improved with intervention dummies for structural breaks, but the overall residual magnitude is small and the fit is adequate.

2.2 Forecasting and Model Comparison

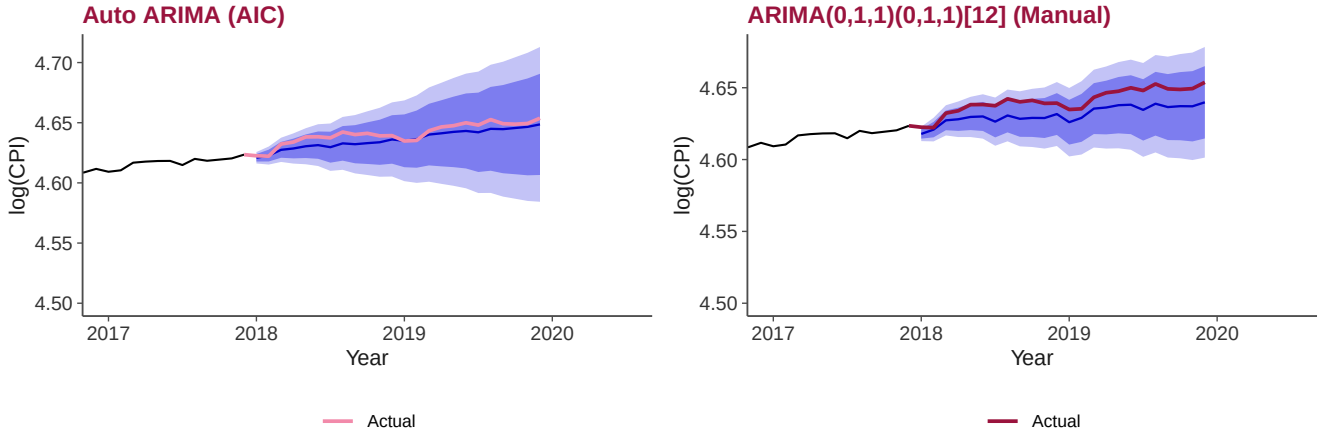

Figure 7: 24-month out-of-sample forecasts (zoomed, 2015-2020). Shaded bands show 80% and 95% prediction intervals.

Table 2: Forecast accuracy comparison.

Model	Set	RMSE	MAE	MAPE	MASE
Auto ARIMA (AIC)	Training set	0.0024	0.0017	0.0499	0.0426
Auto ARIMA (AIC)	Test set	0.0056	0.0050	0.1075	0.1220
ARIMA(0,1,1)(0,1,1)[12]	Training set	0.0025	0.0018	0.0522	0.0441
ARIMA(0,1,1)(0,1,1)[12]	Test set	0.0101	0.0096	0.2071	0.2350

The Auto ARIMA model achieves a test-set RMSE of 0.0056 versus 0.0101 for the airline model, roughly half the forecast error over the 24-month horizon. Both models have a MASE well below 1, outperforming a seasonal naive benchmark. The Auto ARIMA is preferred for forecasting accuracy, though the simpler airline model offers greater parsimony with fewer estimated parameters.